Администрирование локальных сетей

Первоначальное конфигурирование сети

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Цели и задачи работы

Цель лабораторной работы

Провести подготовительную работу по первоначальной настройке коммутаторов сети в Cisco Packet Tracer.

Построение схемы сети

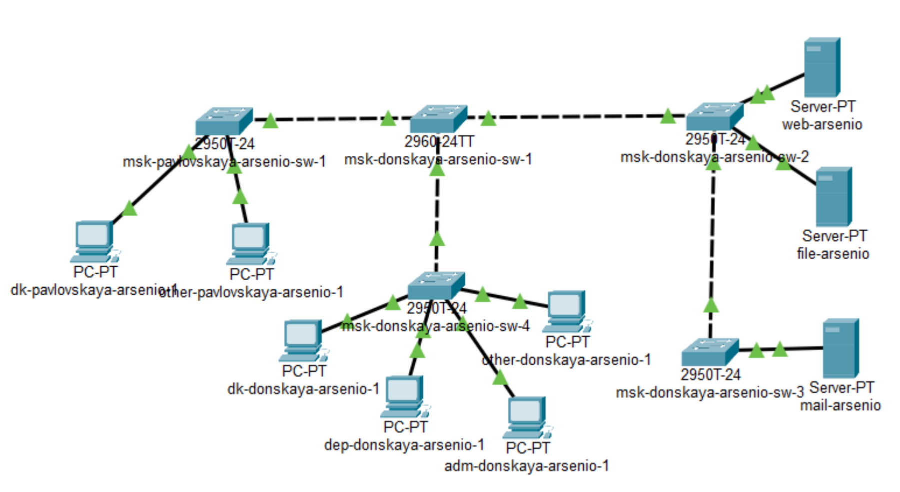

Рис. 1: Размещение коммутаторов и оконечных устройств согласно схеме сети L1

Первоначальная настройка коммутаторов

Настройка msk-donskaya-arsenio-sw-1

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#host msk-donskaya-arsenio-sw-1
msk-donskaya-arsenio-sw-1(config)#int vlan 2
msk-donskaya-arsenio-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to up

msk-donskaya-arsenio-sw-1(config-if)#no sh
msk-donskaya-arsenio-sw-1(config-if)#ip addr 10.128.1.2 255.255.255.0
msk-donskaya-arsenio-sw-1(config-if)#ip default-gate 10.128.1.1
msk-donskaya-arsenio-sw-1(config)#lone vty 0 4
                                    ^
% Invalid input detected at '^' marker.

msk-donskaya-arsenio-sw-1(config)#line vty 0 4
msk-donskaya-arsenio-sw-1(config-line)#pass cisco
msk-donskaya-arsenio-sw-1(config-line)#login
msk-donskaya-arsenio-sw-1(config-line)#line cons 0
msk-donskaya-arsenio-sw-1(config-line)#pass cisco
msk-donskaya-arsenio-sw-1(config-line)#login
msk-donskaya-arsenio-sw-1(config-line)#enab sec cisco
msk-donskaya-arsenio-sw-1(config)#serv pass
msk-donskaya-arsenio-sw-1(config)#user admin priv 1 sec cisco
msk-donskaya-arsenio-sw-1(config)#ip domain-name donskaya.rudn.edu
msk-donskaya-arsenio-sw-1(config)#crypto key gen rsa
The name for the keys will be: msk-donskaya-arsenio-sw-1.donskaya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-arsenio-sw-1(config)#line vty 0 4
*Mar 1 0:8:22.919: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:8:22.919: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-arsenio-sw-1(config-line)#trans in ssh
msk-donskaya-arsenio-sw-1(config-line)#vlan 2
msk-donskaya-arsenio-sw-1(config-vlan)#exit
msk-donskaya-arsenio-sw-1(config)#^Z
msk-donskaya-arsenio-sw-1#
%SYS-5-CONFIG_I: Configured from console by console
```

Настройка msk-donskaya-arsenio-sw-2

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#host msk-donskaya-arsenio-sw-2
msk-donskaya-arsenio-sw-2(config)#int vlan 2
msk-donskaya-arsenio-sw-2(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

msk-donskaya-arsenio-sw-2(config-if)#no sh
msk-donskaya-arsenio-sw-2(config-if)#ip addr 10.128.1.3 255.255.255.0
msk-donskaya-arsenio-sw-2(config-if)#vlan 2
msk-donskaya-arsenio-sw-2(config-vlan)#ip default-gate 10.128.1.1
msk-donskaya-arsenio-sw-2(config)#line vty 0 4
msk-donskaya-arsenio-sw-2(config-line)#pass cisco
msk-donskaya-arsenio-sw-2(config-line)#login
msk-donskaya-arsenio-sw-2(config-line)#line cons 0
msk-donskaya-arsenio-sw-2(config-line)#pass cisco
msk-donskaya-arsenio-sw-2(config-line)#login
msk-donskaya-arsenio-sw-2(config-line)#enab sec cisco
msk-donskaya-arsenio-sw-2(config)#serv pass
msk-donskaya-arsenio-sw-2(config)#user admin priv 1 sec cisco
msk-donskaya-arsenio-sw-2(config)#ip domain-name donskaya.rudn.edu
msk-donskaya-arsenio-sw-2(config)#crypto key gen rsa
The name for the keys will be: msk-donskaya-arsenio-sw-2.donskaya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-arsenio-sw-2(config)#line vty 0 4
*Mar 1 0:11:16.649: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:11:16.649: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-arsenio-sw-2(config-line)#trans in ssh
msk-donskaya-arsenio-sw-2(config-line)#^Z
msk-donskaya-arsenio-sw-2#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-arsenio-sw-2#wr mem
Building configuration...
[OK]
msk-donskaya-arsenio-sw-2#
```

Настройка msk-donskaya-arsenio-sw-3

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#host msk-donskaya-arsenio-sw-3
msk-donskaya-arsenio-sw-3(config)#int vlan 2
msk-donskaya-arsenio-sw-3(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

msk-donskaya-arsenio-sw-3(config-if)#no shut
msk-donskaya-arsenio-sw-3(config-if)#ip addr 10.128.1.4 255.255.255.0
msk-donskaya-arsenio-sw-3(config-if)#vlan 2
msk-donskaya-arsenio-sw-3(config-vlan)#ip default-gate 10.128.1.1
msk-donskaya-arsenio-sw-3(config)#line vty 0 4
msk-donskaya-arsenio-sw-3(config-line)#pass cisco
msk-donskaya-arsenio-sw-3(config-line)#login
msk-donskaya-arsenio-sw-3(config-line)#line cons 0
msk-donskaya-arsenio-sw-3(config-line)#pass cisco
msk-donskaya-arsenio-sw-3(config-line)#login
msk-donskaya-arsenio-sw-3(config-line)#enab sec cisco
msk-donskaya-arsenio-sw-3(config)#serv pass
msk-donskaya-arsenio-sw-3(config)#user admin priv 1 sec cisco
msk-donskaya-arsenio-sw-3(config)#ip domain-name donskaya.rudn.edu
msk-donskaya-arsenio-sw-3(config)#crypto key gen rsa
The name for the keys will be: msk-donskaya-arsenio-sw-3.donskaya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-arsenio-sw-3(config)#line vty 0 4
*Mar 1 0:13:55.242: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:13:55.242: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-arsenio-sw-3(config-line)#trans in ssh
msk-donskaya-arsenio-sw-3(config-line)#^Z
msk-donskaya-arsenio-sw-3#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-arsenio-sw-3#wr mem
Building configuration...
[OK]
msk-donskaya-arsenio-sw-3#
```

Настройка msk-donskaya-arsenio-sw-4

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#host msk-donskaya-arsenio-sw-4
msk-donskaya-arsenio-sw-4(config)#int vlan 2
msk-donskaya-arsenio-sw-4(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

msk-donskaya-arsenio-sw-4(config-if)#no sh
msk-donskaya-arsenio-sw-4(config-if)#ip addr 10.128.1.5 255.255.255.0
msk-donskaya-arsenio-sw-4(config-if)#vlan 2
msk-donskaya-arsenio-sw-4(config-vlan)#ip default-gate 10.128.1.1
msk-donskaya-arsenio-sw-4(config)#line vty 0 4
msk-donskaya-arsenio-sw-4(config-line)#pass cisco
msk-donskaya-arsenio-sw-4(config-line)#login
msk-donskaya-arsenio-sw-4(config-line)#line cons 0
msk-donskaya-arsenio-sw-4(config-line)#pass cisco
msk-donskaya-arsenio-sw-4(config-line)#login
msk-donskaya-arsenio-sw-4(config-line)#enab sec cisco
msk-donskaya-arsenio-sw-4(config)#serv pass
msk-donskaya-arsenio-sw-4(config)#user admin priv 1 sec cisco
msk-donskaya-arsenio-sw-4(config)#ip domain-name donskaya.rudn.edu
msk-donskaya-arsenio-sw-4(config)#crypto key gen rsa
The name for the keys will be: msk-donskaya-arsenio-sw-4.donskaya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-arsenio-sw-4(config)#line vty 0 4
*Mar 1 0:17:15.914: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:17:15.914: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-arsenio-sw-4(config-line)#trans in ssh
msk-donskaya-arsenio-sw-4(config-line)#^Z
msk-donskaya-arsenio-sw-4#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-arsenio-sw-4#wr mem
Building configuration...
[OK]
msk-donskaya-arsenio-sw-4#
```

Настройка msk-pavlovskaya-arsenio-sw-1

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#host msk-pavlovskaya-arsenio-sw-1
msk-pavlovskaya-arsenio-sw-1(config)#int vlan 2
msk-pavlovskaya-arsenio-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to up

msk-pavlovskaya-arsenio-sw-1(config-if)#no sh
msk-pavlovskaya-arsenio-sw-1(config-if)#ip addr 10.128.1.6 255.255.255.0
msk-pavlovskaya-arsenio-sw-1(config-if)#vlan 2
msk-pavlovskaya-arsenio-sw-1(config-vlan)#ip default-gate 10.128.1.1
msk-pavlovskaya-arsenio-sw-1(config)#line vty 0 4
msk-pavlovskaya-arsenio-sw-1(config-line)#pass cisco
msk-pavlovskaya-arsenio-sw-1(config-line)#login
msk-pavlovskaya-arsenio-sw-1(config-line)#line cons 0
msk-pavlovskaya-arsenio-sw-1(config-line)#pass cisco
msk-pavlovskaya-arsenio-sw-1(config-line)#login
msk-pavlovskaya-arsenio-sw-1(config-line)#enab sec cisco
msk-pavlovskaya-arsenio-sw-1(config)#serv pass
msk-pavlovskaya-arsenio-sw-1(config)#user admin priv 1 sec cisco
msk-pavlovskaya-arsenio-sw-1(config)#ip domain-name donskaya.rudn.edu
msk-pavlovskaya-arsenio-sw-1(config)#crypto key gen rsa
The name for the keys will be: msk-pavlovskaya-arsenio-sw-1.donskaya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-pavlovskaya-arsenio-sw-1(config)#line vty 0 4
*Mar 1 0:19:35.659: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:19:35.659: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-pavlovskaya-arsenio-sw-1(config-line)#trans in ssh
msk-pavlovskaya-arsenio-sw-1(config-line)#^Z
msk-pavlovskaya-arsenio-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-pavlovskaya-arsenio-sw-1#wr mem
Building configuration...
[OK]
msk-pavlovskaya-arsenio-sw-1#
```

Выводы

Вывод

В ходе лабораторной работы была построена схема сети L1 в Cisco Packet Tracer.

Была выполнена первоначальная настройка коммутаторов: заданы имена устройств, IP-адреса Vlan2, шлюз по умолчанию, пароли доступа и параметры SSH.

Получены практические навыки базового конфигурирования сетевого оборудования Cisco и сохранения конфигурации в энергонезависимую память.